

CHARCOAT

PASSIVE FIRE PROTECTION

CharCoat IC



Method Statement

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CHARCOAT IC SILICON RUBBER INSULATOR COATING

WORK METHOD STATEMENT

1. CLEANING

- On arrival at the location where the equipment must be coated, the bay in which the equipment insulators are situated must be de-energised, isolated and made safe to work on by the appropriate earthing/grounding methods.
- The surfaces to be coated must be cleaned of all contamination before any further tasks are performed. For loose dust/dirt, cleaning with water and a cloth should suffice. If the contamination is more stubborn, then a non-metallic domestic scouring pad and water should be employed. If there is a greasy residue then cleaning/wiping the surface of the equipment with Toluene will be more than adequate.
- Should there be any deposits or marks that cannot be removed with the scouring pad, the area must be thoroughly rinsed with water to leech out any conductive salts and then the primer and coating applied over the top.
- Alternatively, the use of high-pressure washers with clean water can be considered. Charcoat recommends a nozzle that can expand/fan the water jet. The pressure to be used should be no more than 10 MPa and the distance between nozzle and the insulator surface should be no less than 200mm in order not to harm the equipment surface. The use of a Dry Ice High Pressure system can also be used.

2. BONDING AGENT APPLICATION

- Once clean and dry, the insulator is sprayed with CharCoat IC Primer, to ensure permanent adhesion of the coating to the porcelain or glass surface.
- The bonding agent must be sprayed on as a very thin layer/fine mist using standard spray-painting equipment with the spray gun nozzle having an aperture of 1.5mm. Ideal spraying pressure is 6.5 to 7 Bar (85 to 115 PSI). The primer must just wet the surface without causing any run-off.
- After application, the Primer must be allowed to become touch dry. This takes approximately 15 minutes. It must be noted that the Primer must not be allowed to dry for more than 2 hours before application of the Silicone Rubber Coating.

3. COATING MATERIAL PREPARATION

- A 1 or 5 litre can of the CharCoat IC Silicone rubber coating is to be opened and the small bottle of catalyst provided with the material, added.
- The can is then to be re-closed and shaken for approximately two minutes. This ensures a uniform distribution of the filler materials and catalyst.
- It must be noted that the coating must be applied within 30 minutes of the addition of the catalyst. Every time that coating is decanted from the container, the sprayer must shake the container thoroughly again.

4. COATING APPLICATION

- The CharCoat IC Silicone Rubber Coating is to be applied with standard spray-painting equipment with the spray gun nozzle having an aperture of 2mm and at an operating pressure of 6.5 to 7 Bar (85 to 115 PSI). The coating technique should involve several smooth passes to build up a wet layer thickness of 0,3 to 0,5mm without run-off.
- During spraying, the wet film thickness is to be checked with the Elcometer comb in at least three positions.
- The spraying area must be well ventilated and away from open flames. Personal Protective Equipment/clothing and a respirator mask with appropriate filters must be worn.

5. CURING

- Following coating, the equipment surface will be touch dry within 20 minutes and the High Voltage equipment can be returned to service.
- The full cure of the Silicone Rubber coating will take between 48 and 72 hours after application.

6. CLEANING OF SPRAY EQUIPMENT

- Empty the container of coating material and then pour 150ml of Toluene into the spray gun container, assemble and shake, spray the Toluene into an empty coating container. The disassembled spray gun can also be cleaned with Toluene.

